

If terminal stuck: `sudo killall -9 snort`

SNORPY for rule writing

<https://www.cyb3rs3c.net/>

SNORPY

A Web Based Snort Rule Creator / Maker for Building Simple Snort Rules

alert source port dest port

Rule Message (\ Escape special characters) Class-Type Priority gid

ICMP

ICMP TYPE

ICMP CODE

Data Size

Reference

Threshold Tracking Type TRK BY Count # Seconds

Add Content Match

Add Regex Match

alert icmp any any -> any any (sid:100001; rev:1;)

SNORPY

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alert

WEB Class-Type Priority gid

TCP

PUT HTTP RESPONSE CODE

ACK ☒ SYN ☒ PSH ☒ RST ☒ FIN ☒ URG ☒ ☐ ☐

DIRECTION TCP STATE

> 128

Reference

Threshold Tracking Type TRK BY Count # Seconds

Add Content Match

Add Regex Match

alert tcp any any -> any 80 (msg:"WEB"; content:"PUT"; http_method; flags:ASPRFU; dsize:>128; sid:1000001; rev:1;)

SNORPY

A Web Based Snort Rule Creator / Maker for Building Simple Snort Rules

alert	icmp	any	source port	»	any	dest port	1000001	1
PING detected						Class-Type	Priority	gid

ICMP

ICMP TYPE

ICMP CODE

>

128

Reference

Threshold Tracking Type

TRK BY

Count #

Seconds

Add Content Match

Add Regex Match

alert icmp any any -> any any (msg:"PING detected"; dsize:>128; sid:1000001; rev:1;)

Rule Options: The rule option section contains alert message and information on which part of packet should be inspected to determine if rule action should be taken.

Rule Syntax:

```
<action> <proto> <src_ip> <src_port> -> <dst_ip> <dst_port>
(msg:"<msg>"; sid:<signature>; rev:1;)
```

For custom rule, signature ID starts from 100001, before 100000 reserved for snort.

Rule Actions:

alert : Alert and log the packet
log : log the packet
pass : ignore the packet
drop : block and log the packet
reject : block the packet, log it and send TCP reset
sdrops : block the packet and do not log it

+++++

a) ANY packet detection:

```
$ vim /etc/snort/rules/local.rules
```

```
alert ip any any -> any any (msg: "IP Packet detected"; sid: 100001; rev:1;)
```

```
$ sudo snort -A console -q -i ens33 -c /etc/snort/snort.conf
```

The screenshot shows a terminal window with two panes. The left pane shows the configuration of a snort rule in `/etc/snort/rules/local.rules` and the execution of snort. The right pane shows the output of a ping command from a Windows command prompt.

```
mistOfVON ~
$ sudo vim /etc/snort/rules/local.rules
mistOfVON ~
$ cat /etc/snort/rules/local.rules
alert ip any any -> any any (msg: "IP Packet detected"; sid: 100001; rev:1;)
mistOfVON ~
$ sudo snort -A console -q -i ens33 -c /etc/snort/snort.conf
86/27-01:35:25.943612  [**] [1:100001:1] IP Packet detected [**] [Priority: 0] {ICMP}
 192.168.74.1 -> 192.168.74.135
86/27-01:35:25.944235  [**] [1:100001:1] IP Packet detected [**] [Priority: 0] {ICMP}
 192.168.74.135 -> 192.168.74.1
86/27-01:35:26.946967  [**] [1:100001:1] IP Packet detected [**] [Priority: 0] {ICMP}
 192.168.74.1 -> 192.168.74.135
86/27-01:35:26.947000  [**] [1:100001:1] IP Packet detected [**] [Priority: 0] {ICMP}
 192.168.74.135 -> 192.168.74.1
86/27-01:35:27.963209  [**] [1:100001:1] IP Packet detected [**] [Priority: 0] {ICMP}
 192.168.74.1 -> 192.168.74.135
86/27-01:35:27.963258  [**] [1:100001:1] IP Packet detected [**] [Priority: 0] {ICMP}
 192.168.74.135 -> 192.168.74.1
86/27-01:35:28.978617  [**] [1:100001:1] IP Packet detected [**] [Priority: 0] {ICMP}
 192.168.74.1 -> 192.168.74.135
86/27-01:35:28.978657  [**] [1:100001:1] IP Packet detected [**] [Priority: 0] {ICMP}
 192.168.74.135 -> 192.168.74.1
86/27-01:36:19.693565  [**] [1:100001:1] IP Packet detected [**] [Priority: 0] {UDP}
 192.168.74.2:53 -> 192.168.74.135:33668
86/27-01:36:19.957366  [**] [1:100001:1] IP Packet detected [**] [Priority: 0] {TCP}
 81.189.91.97:80 -> 192.168.74.135:34198
```

```
PS C:\Users\Lenovo> ping 192.168.74.135
Pinging 192.168.74.135 with 32 bytes of data:
Reply from 192.168.74.135: bytes=32 time=1ms TTL=64
Reply from 192.168.74.135: bytes=32 time=1ms TTL=64
Reply from 192.168.74.135: bytes=32 time=1ms TTL=64
Reply from 192.168.74.135: bytes=32 time=1ms TTL=64

Ping statistics for 192.168.74.135:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 1ms, Average = 1ms
PS C:\Users\Lenovo>
```

Snort Rule Writing and Testing

Professional notes covering Snort rule writing, testing, alert modes, logging and community rules.

1. Basic Rule Syntax

```
action protocol source_ip source_port direction destination_ip destination_port (options)
```

- Rule Header
- Rule Options

2. Rule Header

```
alert icmp any any -> $HOME_NET any
```

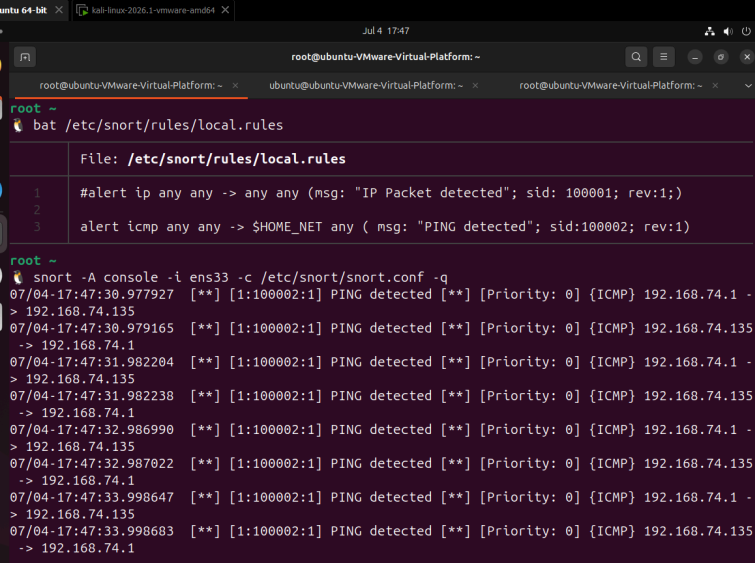
- Action
 - Protocol
 - Source
 - Direction
 - Destination

3. Rule Options

```
(msg:"ICMP Ping Detected"; sid:1001; rev:1;)
```

- msg
- sid
- rev
- content
- flags
- classtype
- priority

```
alert icmp any any -> $HOME_NET any ( msg: "PING detected"; sid:100002; rev:1)
```



The screenshot shows a Kali Linux terminal window with the following content:

```
root ~
bat /etc/snort/rules/local.rules

File: /etc/snort/rules/local.rules

1 #alert ip any any -> any any (msg: "IP Packet detected"; sid: 100001; rev:1;)
2
3 alert icmp any any -> $HOME_NET any ( msg: "PING detected"; sid:100002; rev:1)

root ~
snort -A console -i ens33 -c /etc/snort/snort.conf -q
07/04-17:47:30.977927  [**] [1:100002:1] PING detected [**] [Priority: 0] {ICMP} 192.168.74.1 -
> 192.168.74.135
07/04-17:47:30.979165  [**] [1:100002:1] PING detected [**] [Priority: 0] {ICMP} 192.168.74.135
-> 192.168.74.1
07/04-17:47:31.982204  [**] [1:100002:1] PING detected [**] [Priority: 0] {ICMP} 192.168.74.1 -
> 192.168.74.135
07/04-17:47:31.982238  [**] [1:100002:1] PING detected [**] [Priority: 0] {ICMP} 192.168.74.135
-> 192.168.74.1
07/04-17:47:32.986990  [**] [1:100002:1] PING detected [**] [Priority: 0] {ICMP} 192.168.74.1 -
> 192.168.74.135
07/04-17:47:32.987022  [**] [1:100002:1] PING detected [**] [Priority: 0] {ICMP} 192.168.74.135
-> 192.168.74.1
07/04-17:47:33.998647  [**] [1:100002:1] PING detected [**] [Priority: 0] {ICMP} 192.168.74.1 -
> 192.168.74.135
07/04-17:47:33.998683  [**] [1:100002:1] PING detected [**] [Priority: 0] {ICMP} 192.168.74.135
-> 192.168.74.1
```

On the right side of the terminal, the output of the command `C:\Users\Lenovo>ping 192.168.74.135` is shown:

```
Pinging 192.168.74.135 with 32 bytes of data:
Reply from 192.168.74.135: bytes=32 time=1ms TTL=64
Reply from 192.168.74.135: bytes=32 time=1ms TTL=64
Reply from 192.168.74.135: bytes=32 time=1ms TTL=64
Reply from 192.168.74.135: bytes=32 time=1ms TTL=64

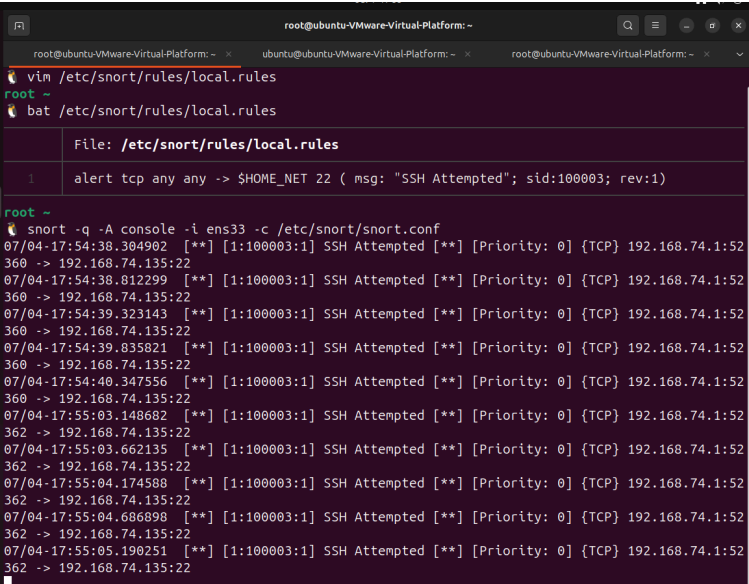
Ping statistics for 192.168.74.135:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\Users\Lenovo>
```

Example Rule

```
alert tcp any any -> $HOME_NET 22 (msg:"SSH Authentication Attempt";
sid:1002; rev:1)
```

- Detects SSH on port 22



The screenshot shows a Kali Linux terminal window with the following content:

```
root ~
vim /etc/snort/rules/local.rules
bat /etc/snort/rules/local.rules

File: /etc/snort/rules/local.rules

1 alert tcp any any -> $HOME_NET 22 ( msg: "SSH Attempted"; sid:100003; rev:1)

root ~
snort -q -A console -i ens33 -c /etc/snort/snort.conf
07/04-17:54:30.304902  [**] [1:100003:1] SSH Attempted [**] [Priority: 0] {TCP} 192.168.74.1:52
360 -> 192.168.74.135:22
07/04-17:54:30.812299  [**] [1:100003:1] SSH Attempted [**] [Priority: 0] {TCP} 192.168.74.1:52
360 -> 192.168.74.135:22
07/04-17:54:39.323143  [**] [1:100003:1] SSH Attempted [**] [Priority: 0] {TCP} 192.168.74.1:52
360 -> 192.168.74.135:22
07/04-17:54:39.835821  [**] [1:100003:1] SSH Attempted [**] [Priority: 0] {TCP} 192.168.74.1:52
360 -> 192.168.74.135:22
07/04-17:54:40.347556  [**] [1:100003:1] SSH Attempted [**] [Priority: 0] {TCP} 192.168.74.1:52
360 -> 192.168.74.135:22
07/04-17:55:03.148682  [**] [1:100003:1] SSH Attempted [**] [Priority: 0] {TCP} 192.168.74.1:52
362 -> 192.168.74.135:22
07/04-17:55:03.662135  [**] [1:100003:1] SSH Attempted [**] [Priority: 0] {TCP} 192.168.74.1:52
362 -> 192.168.74.135:22
07/04-17:55:04.174588  [**] [1:100003:1] SSH Attempted [**] [Priority: 0] {TCP} 192.168.74.1:52
362 -> 192.168.74.135:22
07/04-17:55:04.686898  [**] [1:100003:1] SSH Attempted [**] [Priority: 0] {TCP} 192.168.74.1:52
362 -> 192.168.74.135:22
07/04-17:55:05.190251  [**] [1:100003:1] SSH Attempted [**] [Priority: 0] {TCP} 192.168.74.1:52
362 -> 192.168.74.135:22
```

On the right side of the terminal, the output of the command `C:\Users\Lenovo>ssh 192.168.74.135` is shown:

```
ssh: connect to host 192.168.74.135 port 22: Co
nnection refused

C:\Users\Lenovo>
```

Test Configuration

```
sudo snort -T -i ens33 -c /etc/snort/snort.conf
```

Run Snort

```
sudo snort -q -l /var/log/snort -i ens33 -A console -c /etc/snort/snort.conf
```

- -q Quiet
- -l Log dir
- -A Alert mode

Alert Modes

```
-A console
```

```
-A fast
```

```
-A full
```

```
-A none
```

- Console=Live
- Fast=SIEM
- Full=Verbose
- None=No alerts

Logs

```
/var/log/snort/alert
```

- PCAPs: /var/log/snort/snort.log.*
- Open with Wireshark

Community Rules

```
Snort Community Rules / Cisco Talos
```

- Malware
- CVEs
- Botnets
- Web attacks

Best Practices

- Unique SID
- Increment rev
- Use local.rules
- Run snort -T
- Use -A fast in production

Quick Revision

Topic	Value
Rules	/etc/snort/rules/local.rules
Validate	snort -T
Logs	/var/log/snort
Live Alerts	-A console
SIEM	-A fast